

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**

Number of Questions: **10**

Instructor Name: **Mamani Velasco, Ruben**

Question 1 of 10

Consider the following polynomial function.

$$f(x) = -(x + 1)(x + 2)(x - 1)^2$$

Answer the questions regarding the graph of f .
Then, use this information to graph the function.

(a) Choose the end behavior of the graph of f .

(Blank 1,)

Blank 1, Options

- Falls to the left and rises to the right
- Rises to the left and falls to the right
- Rises to the left and rises to the right
- Falls to the left and falls to the right

(b) List each real zero of f according to the behavior of the graph at the x -axis near that zero. If there is more than one answer, separate them with commas. If there is no answer, write "None".

Zero(s) where the graph crosses the x -axis:

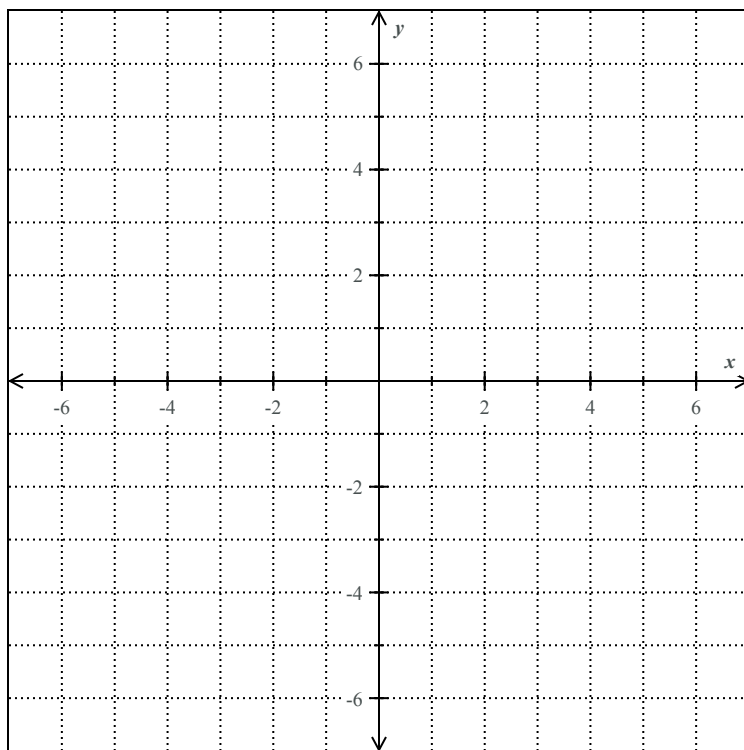
Zero(s) where the graph touches, but does not cross the x -axis:

(c) Find the y -intercept of the graph of f .

y -intercept:

(d) Graph $f(x) = -(x+1)(x+2)(x-1)^2$ by doing the following.

- Plot all points where the graph of f intersects the x -axis or y -axis.
- For each point on the x -axis, select the correct behavior.
- Connect the points following the behaviors.



Question 2 of 10

Write a quadratic function h whose zeros are 1 and 9.

Question 3 of 10

Find a polynomial $f(x)$ of degree 4 that has the following zeros.

8, -9, 6, -3

Leave your answer in factored form.

Question 4 of 10

Divide.

$$(3x^2 + 25x + 46) \div (3x + 7)$$

Your answer should give the quotient and the remainder.

Quotient: _____

Remainder: _____

Question 5 of 10

Divide.

$$(6x^3 + 3x^2 - 20x + 20) \div (2x + 5)$$

Your answer should give the quotient and the remainder.

Quotient:

Remainder:

Question 6 of 10

Use synthetic division to find the quotient and remainder when $3x^3 - 12x^2 + 7$ is divided by $x - 4$ by completing the parts below.

(a) Complete this synthetic division table.

4)

3

-12

0

7

(b) Write your answer in the following form: Quotient + $\frac{\text{Remainder}}{x - 4}$.

$3x^3 - 12x^2 + 7$

$x - 4$

=

+

$x - 4$

Question 7 of 10

For the polynomial below, -1 is a zero.

$$h(x) = x^3 - 3x^2 - 2x + 2$$

Express $h(x)$ as a product of linear factors.

Question 8 of 10

Use the rational zeros theorem to list all possible rational zeros of the following.

$$g(x) = -8x^2 - 3x - 4 - 9x^3 + 8x^4$$

Be sure that no value in your list appears more than once.

Question 9 of 10

Multiply.

$$[x - (-4 + 5i)][x - (-4 - 5i)]$$

Note that these expressions contain complex numbers.
Simplify your answer as much as possible.

Question 10 of 10

Find a polynomial $f(x)$ of degree 4 with real coefficients and the following zeros.

$$4, -3, -2 + i$$

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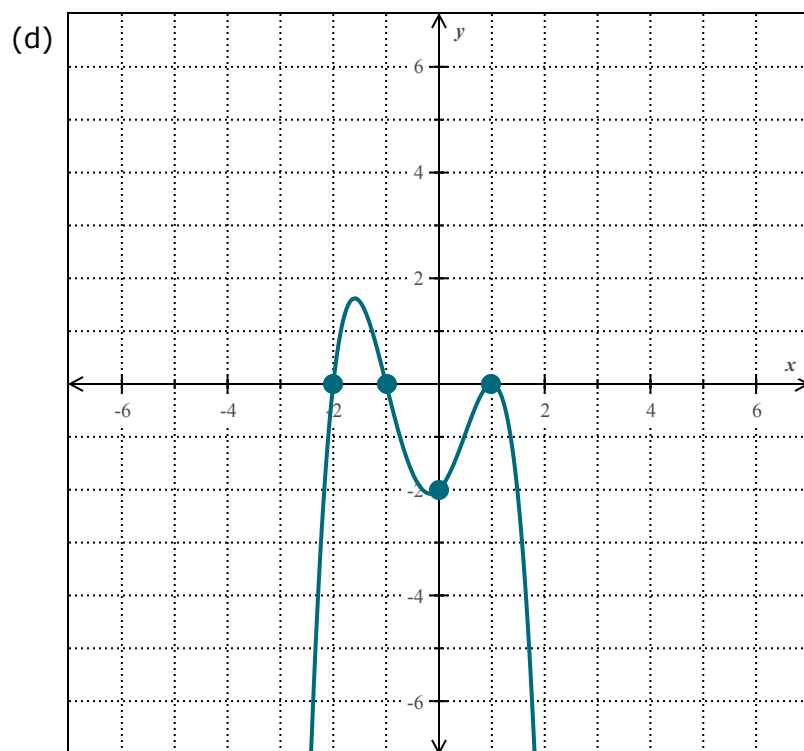
Question 1 of 10

(a) Falls to the left and falls to the right

(b) Zero(s) where the graph crosses the x -axis: $-1, -2$

Zero(s) where the graph touches, but does not cross the x -axis: 1

(c) y -intercept: -2



Question 2 of 10

$$h(x) = (x - 1)(x - 9)$$

Question 3 of 10

$$f(x) = (x - 8)(x + 9)(x - 6)(x + 3)$$

Question 4 of 10Quotient: $x + 6$

Remainder: 4

Question 5 of 10Quotient: $3x^2 - 6x + 5$ Remainder: -5 **Question 6 of 10**

(a) Complete this synthetic division table.

$$\begin{array}{r|rrrr}
 4 & 3 & -12 & 0 & 7 \\
 & & 12 & 0 & 0 \\
 \hline
 & 3 & 0 & 0 & 7
 \end{array}$$

(b) Write your answer in the following form: Quotient + $\frac{\text{Remainder}}{x - 4}$.

$$\frac{3x^3 - 12x^2 + 7}{x - 4} = 3x^2 + \frac{7}{x - 4}$$

Question 7 of 10

$$h(x) = (x + 1)(x - (2 + \sqrt{2}))(x - (2 - \sqrt{2}))$$

Question 8 of 10

$$1, -1, 2, -2, 4, -4, \frac{1}{2}, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{8}, -\frac{1}{8},$$

or in shorthand

$$\pm 1, \pm 2, \pm 4, \pm \frac{1}{2}, \pm \frac{1}{4}, \pm \frac{1}{8}.$$

Question 9 of 10

$$x^2 + 8x + 41$$

Question 10 of 10

$$f(x) = x^4 + 3x^3 - 11x^2 - 53x - 60$$